



Joint RTG Workshop

16-18 September 2024

LinderHof Erfurt / Azmannsdorfer Str. 27
99198 Erfurt



Monday (16 September)		^{1,2,3}H	BENCH
10:00 – 11:30	Invited presentation	DRs	
11:30 – 12:15	Lunch	DRs	
13:00 – 14:30	City tour / meet at Anger Lutherdenkmal	DRs	
15:30 – 16:00	Coffee break	PIs, DRs	
16:00 – 18:00	Progress reports (5 ^{1,2,3} H)	PIs, DRs	
18:00 – 19:00	Dinner	PIs, DRs	
19:00 – 21:00	Poster session Informal discussions	PIs, DRs	
Tuesday (17 September)		^{1,2,3}H	BENCH
09:00 – 10:30	^{1,2,3} H Annual Report	PIs, DRs	
10:30 – 11:00	Coffee break	PIs, DRs	
11:00 – 12:00	^{1,2,3} H PI Meeting / ^{1,2,3} H DR Forum	PIs, DRs	
	<i>Arrival of BENCH members</i>		
12:00 – 13:30	Lunch	PIs, DRs	PIs, DRs
13:30 – 14:10	Welcome / Presentation of RTGs	PIs, DRs	PIs, DRs
14:10 – 15:30	Progress reports (3 BENCH + 1 ^{1,2,3} H)	PIs, DRs	PIs, DRs
15:30 – 16:00	Coffee break	PIs, DRs	PIs, DRs
16:00 – 17:40	Progress reports (3 BENCH + 2 ^{1,2,3} H)	PIs, DRs	PIs, DRs
	<i>Departure of ^{1,2,3}H members not participating in joint workshop</i>		
18:00 – 19:30	Dinner	PIs, DRs	PIs, DRs
19:30 – 21:00	Poster session / Informal discussions	PIs, DRs	PIs, DRs
Wednesday (18 September)		^{1,2,3}H	BENCH
09:00 – 10:40	Progress reports (2 BENCH + 3 ^{1,2,3} H)	PIs, DRs	PIs, DRs
10:40 – 11:00	Coffee break	PIs, DRs	PIs, DRs
11:00 – 12:00	Progress reports (1 BENCH + 2 ^{1,2,3} H)	PIs, DRs	PIs, DRs
12:00 – 13:30	Lunch	PIs, DRs	PIs, DRs
13:30 – 15:30	Discussion of RTG programmes	PIs, DRs	PIs, DRs
15:30 – 16:00	Coffee break	PIs, DRs	PIs, DRs
16:00 – 16:30	Concluding Remarks	PIs, DRs	PIs, DRs
	<i>Departure of PIs</i>		
17:30 – 19:00	Dinner	DRs, (PIs)	DRs, (PIs)
19:00 –	DR-only night	DRs	DRs
Thursday (19 September)		^{1,2,3}H	BENCH
	Departure	DRs, (PIs)	DRs, (PIs)

Programme details

Monday	Progress reports (12+8 minutes)
16:00	Lab-on-a-chip detection of T-species: Evaluating mixing methods for emulsification ^{1,2,3} H / P11 / Jantje Bäcker
16:20	Tritium separation from gaseous hydrogen isotopologue mixtures ^{1,2,3} H / P12 / Alexandra Becker
16:40	Novel photocatalytic pathway for selective activation of α-C–H bonds for a direct late-stage deuteration of alcohols ^{1,2,3} H / P9 / Jannik Sälker
17:00	Selective electrochemical reduction of ketones using a cathode-shielding electrolyte ^{1,2,3} H / P10 / Jonathan Lebelt
17:20	Probing the temperature of single quantum dots confined in a nanoparticle ion trap by fluorescence thermometry ^{1,2,3} H / P1a / Sophia Leippe
Tuesday	Welcome / Presentation of RTGs
13:30	RTG 2721: Hydrogen Isotopes ^{1,2,3} H / Knut Asmis (spokesperson)
13:50	RTG 2455: Benchmark experiments for numerical quantum chemistry BENCh / Martin Suhm (deputy spokesperson)
Tuesday	Progress reports (12+8 minutes)
14:10	Thermodynamics and kinetics of proton-coupled electron transfer reactions involving a dithiol/disulphide redox switch BENCh / Pierre Kachel
14:30	Delta learning of nuclear quantum effects in molec. dynamics simulations ^{1,2,3} H / P15 / Rhyan Barrett
14:50	Into the Reactionverse: Exploring Algorithms for Minimizing Energy Pathways BENCh / Maike Mücke
15:10	Towards metalonitrenyl radical cations BENCh / Nadeshda Shcherbina
15:30	Coffee break
16:00	Investigating the interaction of stable organic radicals and halogenated alcohols BENCh / Elisabeth Sennert
16:20	In-situ electron paramagnetic resonance investigation of isotope-selective breathing in MIL-53 during dihydrogen adsorption ^{1,2,3} H / P5 / Muhammad Fernadi Lukman
16:40	Proton Dynamic and pH effects on Organic Photoswitches BENCh / Radek Tovtik
17:00	Chemical bonding in excited states ^{1,2,3} H / P14a / Florian Kreuter
17:20	A μ-1,1-Nitroso Dicopper complex with unusual electronic structure and its one electron reduction triggering NO release BENCh / Disha Bhattacharya

Wednesday	Progress reports (12+8 minutes)
09:00	Hydrogen isotope effects on chemical bonding in boron cluster complexes ^{1,2,3} H / P14 / Raza Ullah Khan
09:20	Deeper into the quantum realm - low-order scaling electronic/proton structure theory BENCh / Lukas Hasecke
09:40	Investigation of electrochemically driven hydrogen isotope penetration through thin membranes ^{1,2,3} H / P7 / Robert Temme
10:00	Microhydration of Tertiary Amines: Resonances and How to Remove them by Isotope Substitution BENCh / Eaindra Lwin
10:20	Gas phase vibrational spectroscopy of microsolvated hexafluorophosphate anion: solute-solvent vs. solvent-solvent interactions ^{1,2,3} H / P13 / Milena Barp
10:40	<i>Coffee break</i>
11:00	Hydrogen adsorption on isolated clusters ^{1,2,3} H / P1 / Shabnam Haque
11:20	One Ring to bind them all: The dimers of H₂, HD and D₂ with benzonitrile investigated by microwave spectroscopy BENCh / / Beppo Hartwig
11:40	Rational design of materials for adsorptive He isotope separation ^{1,2,3} H / P2 / Elvira Dongmo

Wednesday	Discussion of RTG programmes			
13:00	Selection of two discussion topics and formation of 4 groups, two for each topic (see suggestions below)			
	Group 1	Group 2	Group 3	Group 4
	Discussion of topic A	Discussion of topic A	Discussion of topic B	Discussion of topic B
13:15	Presentation results A			
13:25	Discussion All of topic A			
13:35	Presentation results B			
13:45	Discussion All of topic B			
14:00-15:00	Formation of new groups and analogous discussion and presentation of topics C and D			

Topic suggestions

1. What are the most efficient career development measures?
2. Do's and Don't's of supervision strategies
3. What is the added value of an RTG?
4. How to deal with RTG framework constraints and time pressure?

Other potential topics

1. Recruiting – how to find qualified candidates in today's job market?
2. research data management
3. sustainability
4. equity and diversity

Joint Workshop Participants

^{1,2,3}H members participating in the joint workshop stay Tuesday night. DRs also stay Wednesday night, while PIs depart on Wednesday afternoon (or optionally stay for dinner and depart on Thursday morning).

^{1,2,3} H		BENCH	
PIs	DRs/postdocs	PIs	DRs/postdocs
Bernd Abel	Robert Temme	Marina Bennati	
Knut Asmis	Milena Barp Björn Bastian (from 17) Shabnam Haque Jiaye Jin	Ricardo Mata (18 Sep only)	Maike Mücke Lukas Hasecke
Thomas Heine (may leave early 18 Sep)	Elvira Dongmo	Daniel Obenchain	Beppo Hartwig
Harald Krautscheid	Sibo Chetry	Sven Schneider (18 Sep only)	Sandipan Maji Nadeshda Shcherbina
Agnieszka Kuc	İsmail Eren	Martin Suhm	Eaindra Moh Moh Lwin Elisabeth Sennert
Ralf Tonner-Zech	Raza Ullah Khan Florian Kreuter		Disha Bhattacharya (AK Meyer)
	Rhyan Barrett (AK Westermayr)		Pierre Pascal Kachel (AK Siewert)
	M. F. Lukman (AK Pöppl)		Radek Tovtik (AK Simeth-Crespi)

^{1,2,3}H Members Participating only in the ^{1,2,3}H Annual Workshop

^{1,2,3}H members not participating in the joint workshop leave on Tuesday (or optionally stay for dinner and leave on Wednesday morning).

PIs	DRs/postdocs
	Sophia Leippe (AK Asmis)
Detlev Belder	Jantje Bäcker (+2) Sanjay Lama
Cornelius Fischer	Alexandra Becker Holger Lippold
Roger Gläser	Konstantin Marcinowski David Poppitz
Tanja Gulder	Jonathan Lebelt
Andreas Pöppl	
	Robert Schiewe (NFG Warneke)
Julia Westermayr	Alexander Feige
Kirsten Zeitler	Jannik Sälker